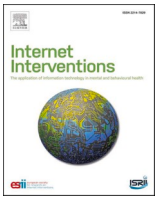
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# MyADHD: A therapist-guided internet-delivered intervention for adult ADHD — Results from a single-armed open clinical trial in routine care

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A B S T R A C T

*Background:* Adults seeking treatment for the syndrome of ADHD often face barriers to accessing evidence-based care. Digital interventions may help address these challenges, but their effectiveness in routine clinical settings remains underexplored.
*Objective:* This study evaluated pre–post change in ADHD symptoms and quality of life following a therapist-guided internet-delivered intervention for adults with ADHD in routine care. A secondary aim was to examine demographic, contextual, and clinical predictors of treatment response.
*Methods:* In an open, single-arm trial, 228 adults with ADHD received a 7–10 week therapist-guided intervention. ADHD symptoms (Adult ADHD Self-Report Scale; ASRS) and quality of life (Adult ADHD Quality of Life questionnaire; AAQoL) were assessed at baseline, mid-treatment, and post-treatment. Random intercept, fixed slope linear mixed models were estimated to examine change over time and impact of potential predictors.
*Results:* Participants showed moderate improvements in ADHD symptoms ($d = -0.47$) and quality of life ($d = 0.45$). Inattention and productivity domains improved most. Reliable change was observed in 23.9% of completers for ADHD symptoms and 31.0% for quality of life. No demographic, contextual, or clinical variables significantly predicted treatment response.
*Conclusions:* In this open, single-arm study conducted in routine clinical care, therapist-guided internet-delivered treatment was associated with moderate improvements in ADHD symptoms and quality of life among adults with ADHD. Findings were comparable across baseline levels of comorbidity, treatment expectation and route to care. Given the absence of a control group, findings should be interpreted cautiously, and causal inferences cannot be drawn. Replication in adequately powered randomized controlled trials is needed to determine the intervention's efficacy and to clarify for whom and under what conditions it is most effective.

## 1. Introduction

ADHD is often conceptualized as a neurodevelopmental disorder with originates in early childhood and continues into adulthood (American Psychiatric Association, 2022; Franke et al., 2018). It is estimated that about 5% of children and 2–3% of adults meet the criteria for an ADHD diagnosis (Faraone et al., 2021; Polanczyk et al., 2007; Song et al., 2021). ADHD is associated with a wide range of negative life outcomes (Barkley, 2002; Franke et al., 2018), including increased rates of unemployment (Samosh et al., 2024), high-school and university discontinuation or dropout (Fried et al., 2016; Müller et al., 2024), higher utilization of psychiatric and somatic healthcare services (Instanes et al., 2018; Sobanski, 2006), incarceration, and a high prevalence of comorbid psychiatric disorders (Sobanski, 2006). Common comorbidities include substance use disorders, anxiety, depression, and personality pathology (Choi et al., 2022; Sobanski, 2006).

For both child and adult populations, pharmacological interventions are the primary recommended treatment (Helsedirektoratet, 2016; National Institute for Health and Care Excellence, 2018). Stimulants often being the first choice, followed by non-stimulant alternatives. These interventions have been generally found to be highly effective in reducing symptoms and improving functioning (Faraone et al., 2021; Groom and Cortese, 2022). Furthermore, there are indications that successful pharmacological treatment may facilitate engagement with other non-pharmacological treatment options (Faraone et al., 2021; Li and Zhang, 2024). Despite their reported effectiveness, adherence to pharmacological treatment is inconsistent, particularly among adults (Adler and Nierenberg, 2010; Charach and Fernandez, 2013). In fact, Brikell et al. (2024), a landmark retrospective study of population-based data from eight countries found that more than half of adult (between 61 and 52%) and adolescent (53%) patients discontinued pharmacological treatment within one year. Discontinuation may be influenced by a

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range of factors, including side effects, lack of perceived benefit, or difficulty maintaining long-term treatment adherence more generally (Adler and Nierenberg, 2010; Charach and Fernandez, 2013). Furthermore, stimulant treatments are associated with a range of adverse effects, including sleep disturbance, appetite suppression, anxiety, mood lability, cardiovascular changes such as increased blood pressure, and (albeit more rarely) psychotic or movement-related symptoms, particularly in individuals with comorbid psychiatric or somatic comorbidities (Graham et al., 2011; Graham and Coghill, 2008; Silczuk et al., 2025).

Non-pharmacological interventions have also been found effective in addressing both ADHD symptoms and improving functioning and quality of life (Faraone et al., 2021; Fullen et al., 2020; Li and Zhang, 2024; Nimmo-Smith et al., 2020). Cognitive behavioral therapy (CBT)-based interventions are particularly well-documented in the research literature (Fullen et al., 2020). These treatments can serve as a supplement to pharmacological interventions or as an alternative for individuals who discontinue or prefer not to use medication (Faraone et al., 2021; Li and Zhang, 2024).

Despite the documented effectiveness of both pharmacological and non-pharmacological treatments, access to targeted interventions can be highly variable — especially for adults (Franke et al., 2018; Solberg et al., 2019). In fact, access to care specifically addressing ADHD related issues is often limited due to scarce resources and capacity, and adult ADHD patients express feeling de-prioritized in the health care system (Stivala, 2021). This is particularly concerning given the increasing number of people seeking treatment for ADHD (Martin et al., 2025; University of Utah Health, 2024), and given the potential burden these patients may experience in their daily lives (Oscarsson et al., 2022). In Norway, for example, adults with ADHD represent a growing proportion of the treatment-seeking population, with certain hospitals reporting neurodevelopmental disorders like ADHD taking up 8.5% of all outpatient sessions in adult care (Helse Sør-Øst RHF, 2024a). Furthermore, many treatment seeking adults are rejected, with one Norwegian hospital reporting that up to 25% of all rejected referrals are ADHD related referrals with insufficient documentation of impairment (Helse Sør-Øst RHF, 2024b). Adults seeking out assessment and treatment for ADHD have also been highlighted by one of the four major healthcare regions as a growing patient group who is often rejected in specialist care services due to insufficiently documented functional impairment and limited capacity (Helse Sør-Øst RHF, 2024c).

In response to growing demands, digital mental healthcare interventions represent a promising solution to expand access to treatment. Whether administered as self-help applications or guided digital intervention programs, digital interventions hold great promise for addressing challenges of capacity and variable access. Research into the acceptability, utilization, and effectiveness of digital interventions have shown consistently good results across a wide range of mental disorders (Rogers et al., 2017; Titov et al., 2018; Vernmark et al., 2024; Yogarajah et al., 2020). Digital interventions address the problem of access on two levels (Anser et al., 2025; Conrad, 2024; Mwogosi, 2025): Firstly, it breaks down the barrier of geographic distances and localized access; Second, the capacity of individual healthcare professionals is greatly increased as much of the therapeutic content is predetermined and offered automatically. Digital interventions for patients with ADHD have so far shown promise (see for instance Gabarron et al., 2025 for a comprehensive meta-review; or Oscarsson et al., 2025 for a recent example).

In Norway, a self-guided internet-delivered intervention for adults with ADHD (MyADHD) has previously been developed to support symptom management and everyday functioning (Flobak et al., 2021; Nordby et al., 2021). This self-guided intervention was found to be effective in a community sample of individuals with self-reported ADHD (Kenter et al., 2023). While these findings are promising, there is some evidence suggesting that adults with ADHD may benefit more from guided rather than self-guided approaches (Gabarron et al., 2025). Guided interventions tend to improve adherence and treatment engagement, factors that are particularly relevant for individuals with ADHD, who often experience difficulties with sustained attention, organization, and self-regulation. Moreover, therapist or coach guidance allows for individualized feedback and adaptive pacing. The community sample used in the initial study (Kenter et al., 2023) likely consisted of highly motivated individuals, who may have been better able to engage independently with the digital content. In contrast, patients in outpatient mental health clinics typically present with clinically confirmed ADHD, higher symptom burden, and greater functional impairments factors that can make it more challenging to adhere to self-guided intervention. To better meet the needs of clinical ADHD populations, an updated version of the therapist-guided intervention MyADHD has been developed by adapting the original self-help intervention by Kenter et al. (2023) to include asynchronous therapist guidance. Further research is needed to determine the effectiveness of this guided intervention when implemented in real-world outpatient settings. Furthermore, while previous studies have demonstrated the feasibility, acceptability, and effectiveness of the intervention (Kenter et al., 2023; Nordby et al., 2021), little is known about which, if any, factors predict treatment response for this program.

## 2. The current study

The current study is an open, single-arm intervention study conducted in a routine care setting at an eTreatment clinic specialized in providing internet-delivered interventions. The primary aim is to evaluate pre-post change in ADHD symptom severity and quality of life following a therapist-guided internet-delivered intervention for adults with ADHD. A secondary aim is to explore the impact of potential predictors of treatment response, including demographic characteristics, contextual factors, and psychiatric comorbidity. To address these aims, the study poses the following research questions:

RQ1: What is the overall effect of a therapist-guided internet-delivered intervention for adult ADHD on ADHD symptom severity and quality of life, when administered to a clinical sample in a routine care setting?

RQ2: To what extent do demographic (e.g., age, sex, employment status, education level), contextual (e.g., treatment expectations, referral pathway, medication status), and clinical (e.g., depression, anxiety, personality pathology) factors predict treatment response to this therapist-guided internet-delivered intervention?

## 3. Methods

### 3.1. Design, participants and procedure

The current study was a one-armed open trial conducted in routine care at the eTreatment Clinic, Department of Psychiatry, Haukeland University Hospital, Bergen Norway. Participants were treatment-seeking adults with a self-reported ADHD diagnosis, who at the time of the study qualified for specialist psychiatric care due to issues related to ADHD. Participants were recruited between January 2024 and June 2025.

Study participation was predicated on the subject being ≥18 years old, already having an ADHD diagnosis, having access to a computer/smartphone and the internet, and speaking Norwegian fluently. Furthermore, participation was barred for people in need of immediate psychiatric treatment for other ongoing mental illness such as e.g., bipolar disorder or severe suicidality.

As the current study was carried out in routine care, participation was further predicated on subjects qualifying for the right to specialist mental healthcare in accordance with Norwegian clinical practice standards. Participants also had to live within the greater Bergen municipal area served by the eTreatment Clinic at Haukeland University Hospital. These and all other inclusion criteria were evaluated by the

treating clinician upon referral to the treatment program, in accordance with routine practice at the eTreatment Clinic.

Prospective participants were referred to the treatment either by their GP, a mental healthcare professional outside the eTreatment clinic, or themselves — through an online application portal managed by the Norwegian healthcare authorities (Helse Norge).

### 3.2. Intervention

The intervention (MyADHD) is a therapist-guided internet-delivered, structured treatment for adults with ADHD, based on modified elements from Cognitive Behavioral Therapy, Dialectical Behavioral Therapy and Goal Management Training. The intervention was originally developed as a self-help intervention but has recently been revised to include asynchronous therapist guidance. Each patient is assigned a therapist who oversees their treatment, monitors their progress through the intervention program, and answers any questions or messages the patient may send. Therapists typically engage directly with their patients once a week through asynchronous messages sent through the platform. The therapist's role is to guide the patient through the treatment content, motivate them to complete the intervention and answer questions the patient may have about the intervention content. The therapeutic content is largely unchanged from previous iterations (see Flobak et al., 2021; Kenter et al., 2023; Nordby et al., 2021).

In the current study, the treatment unfolded as a structured, yet flexible program delivered over approximately 7–10 weeks, during which patients progressed through seven self-contained modules addressing key therapeutic topics—Treatment Introduction and goal setting, Mindful Awareness, Inhibition Training, Emotional Regulation, Planning/Organizing, Self-Acceptance, and a Treatment Summary. Patients were oriented to the program and its rationale in the initial module, after which subsequent modules were unlocked sequentially on a weekly basis, contingent on completion of the preceding module, allowing for individual differences in pacing within the overall treatment window. Each module integrated case descriptions, psychoeducational material, and a range of therapeutic techniques and exercises, with content delivered through a combination of text, audio, and video. Throughout the program, patients were expected to engage with the material and practice the introduced strategies in their everyday lives, while asynchronous therapist guidance supported engagement, clarified content, and encouraged the application of program content to each patient's individual challenges.

### 3.3. Measures

#### 3.3.1. Primary and secondary outcome measures

Self-reported *ADHD symptom severity* and self-reported *quality of life* were both measured at pre-treatment (T0), after reaching the midpoint of treatment upon completion of the third treatment module (T1), and after completing the sixth treatment module (T2). Participants who discontinued treatment or did not complete a pending assessment were not administered further assessments, owing to constraint inherent to the naturalistic study context.

The Adult ADHD Self-Rating Scale (ASRS) was used to assess self-reported ADHD symptom severity. The instrument consists of 18 items corresponding to the symptoms of ADHD described in the diagnostic manual (DSM-5). Items are grouped into the two sub-scales of inattention and hyperactivity/impulsivity, each comprised of 9 items. Items are rated on a five-point Likert scale to indicate the subject's frequency of experiencing symptoms, ranging from 0 (indicating "Never") to 4 (indicating "Very often"). Both total score and sub-scale scores were included as primary and secondary outcomes respectively. Cronbach's alpha for the total score in the current sample was 0.81.

The Adult ADHD Quality of Life questionnaire (AAQoL) was used to measure self-reported quality of life. The instrument consists of 29 items assessing health-related quality of life over the past two weeks among adults with ADHD. Items are grouped into four sub-scales of Life Productivity (11 items), Psychological Health (6 items), Life Outlook (7 items), Relationships (5 items). Items are rated on a five-point Likert scale ranging from 1 to 5, indicating either degree of experienced difficulty (with 1 indicating "Not at all" and 5 indicating "Extreme") or frequency of experience (with 1 indicating "Never" and 5 indicating "Very often"). Both total score and sub-scale scores were included as primary and secondary outcomes respectively. Cronbach's alpha for the total score in the current sample was 0.90.

### 3.4. Baseline predictors

In addition to the primary and secondary outcome measures, additional information was recorded at baseline to explore the impact of Psychiatric Comorbidity, Contextual Factors and Demographics.

#### 3.4.1. Comorbidity

Depressive symptoms were assessed using the Patient Health Questionnaire (PHQ-9; Löwe et al., 2004), which includes nine items scored 0 ("not at all") to 3 ("nearly every day"), with total scores ranging 0–27. For the purposes of analysis, participants were dichotomized into "depressed" vs. "not depressed" at baseline, with depression defined as a total score $\geq$ 10 *and* a score $\geq$ 2 on at least one of the first two items (reflecting core symptoms of low mood or anhedonia). See Kroenke et al. (2001) for documentation of these cut-off values. Cronbach's alpha for the total score in the current sample was 0.84.

Anxiety was assessed with the Generalized Anxiety Disorder scale (GAD-7; Spitzer et al., 2006; Kroenke et al., 2007), a seven-item scale scored 0–3, with total scores ranging 0–21. Participants were categorized as having an anxiety disorder if they scored $\geq$10 at baseline. See O'Connor et al. (2023) for documentation for this cut-off value. Cronbach's alpha for the total score in the current sample was 0.85.

Personality dysfunction was assessed using the PDS-ICD-11 (Bach et al., 2021, 2023; Brown & Sellbom, 2023), a 14-item self-report measure aligned with ICD-11 diagnostic criteria for personality disorders. For analytic purposes, participants were classified as having personality pathology if they scored $\geq$16 on the total scale *and* $\geq$2 on item 14 (reflecting the degree of impairment to daily life). See (Bach et al., 2023) for documentation of these cut-off values. Cronbach's alpha for the total score in the current sample was 0.80.

#### 3.4.2. Contextual factors

These included self-reported variables indicating whether participants were currently using *medication* to manage their ADHD (Yes/No), the *referral pathway* of the patient (referred by their GP or a mental healthcare professional outside the eTreatment Clinic/self-referred), and the *patients' rating of expectations* of benefit from the treatment (on a scale of 1 to 10, with 1 indicating no expectation of benefiting from treatment and 10 indicating strong expectation of benefiting from treatment).

#### 3.4.3. Demographic factors

These included participants' self-reported *age*, *sex*, whether they were employed or enrolled as students, and whether or not they had attained a university level *education*.

### 3.5. Statistical analysis

Two sets of random intercept fixed slopes models were estimated for the current study, one set to evaluate overall change in primary and secondary outcome measures across treatment and one set to assess the impact of predictors on change in the two primary outcome measures.

To assess change over time across the treatment period, separate models were estimated for each outcome measure using two dummy coded contrasts: one representing change from pre- to mid-treatment (T0 = 0, T1 = 1, T2 = 1) and one representing change from mid- to

post-treatment (T0 = 0, T1 = 0, T2 = 1). Together, these contrasts allow separate estimates of change across the two treatment phases, with pre-treatment baseline serving as the model intercept. Effect sizes were calculated for pre-mid and pre-post change based on differences between timepoints using the following formula by Morris and DeShon (2002): $d = \frac{M_{Post} - M_{Pre}}{SD_{Pre}}$. Reliable change was calculated for participants who had completed pre-and post-assessments using the following formula by Jacobson and Truax (1991): $RC = \frac{X_2 - X_1}{Sdiff} \rightarrow RCI = \pm 1.96 * SD_{Pre}\left(\sqrt{2*(1-r)}\right)$.

To assess the impact of predictors on change over time across the treatment period, separate models were estimated for both primary outcome measures, with each model exploring the effect of a single predictor on the change in the given outcome variable. For these models, time was coded as a three-level factor variable (T0 = 0, T1 = 1, T2 = 2). Models included the fixed effect of time, the fixed effect of the predictor, and an interaction term for time × predictor.

All analyses were carried out in STATA version 18.0. Linear mixed models were estimated using restricted maximum likelihood (REML) with Satterthwaite's approximation for degrees of freedom. Missing data were handled under the missing at random (MAR) assumption inherent to linear mixed models. While the MAR assumption cannot be empirically tested, its plausibility was explored by summarizing common dropout patterns and comparing baseline characteristics across these patterns to evaluate the presence of systematic differences.

Additionally, for the two primary outcome variables of ASRS and AAQoL total scores, two additional rounds of linear mixed models assessing change over time were estimated for (a) study completers only and (b) all participants with missing values imputed using multiple imputation — where 10 imputed datasets were generated using multiple imputation for monotone missingness. Imputation models included the categorical time variable, baseline ASRS and AAQoL scores, treatment expectations and age; As well as dichotomous indicators of comorbid anxiety, depression, and personality pathology, medication status, education, and sex. These additional analyses were conducted as part of the sensitivity analyses to examine the potential impact of attrition and missing data assumptions on estimates of change over time. Results from these models were compared with those from the primary analysis, which included all available data without imputation. Similar findings across repeated analyses were interpreted as suggesting limited sensitivity of results to missing data and attrition.

## 4. Results

### 4.1. Sample

A total of 234 participants were enrolled in the study between January 2024 and June 2025. Six of these participants had discontinued their activity in the program before completing the baseline assessment and were therefore excluded from later analysis. The remaining 228 participants were predominantly female (73.1%), with a mean age of 31.9; most participants (68.5%) were employed (51.8%) or enrolled in university (16.7%) at the time of the study and a sizable portion (43.4%) reported having previously completed some form of higher-level education. Most participants (62.3%) were using medication for their ADHD at the time of the study. Being referred to the treatment program by either their GP (11%) or a mental health professional outside the eTreatment clinic (67.5%) was far more common than being self-referred (21.5%).

Table 1 shows baseline ADHD symptoms, quality of life measures, treatment expectation and psychiatric comorbidity for the full sample and various dropout patterns. As can be seen from the table, the average ASRS and AAQoL scores were within clinician range (see Adler et al., 2019; Brod et al., 2015; NovoPsych, 2025). Furthermore, comorbidity was widespread with most participants (64.9%) qualifying for at least one comorbid disorder. Average treatment expectation was also somewhat low among participants of the current study, compared to previous studies of treatment expectation for digital interventions (see for instance Pontén et al., 2024).

### 4.2. Missing data

Across demographic variables, some minor data discrepancies were identified and handled. A total of two participants had entered negative values for their age and two additional participants had entered ages below 18, which would be impossible given the way inclusion to the study was handled by the eTreatment clinic. Similarly, a total of four participants reported having been diagnosed with ADHD before the age of 4, which would be out of line with current clinical guidelines (Wolraich et al., 2019) and highly unlikely from a Norwegian clinical

**Table 1**
Overview of baseline symptoms, quality of life, comorbidity, demographics and contextual factors across full sample and dropout patterns.

| Continuous variables | Total sample (N = 228) | | Early dropout (n = 61) | | Late dropout (n = 54) | | Completers (n = 113) | | Early/completers | | Late/completers | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | M | (SD) | M | (SD) | M | (SD) | M | (SD) | *p* | *t* | *p* | *t* |
| Age | 31.86 | (9.70) | 29.41 | (7.82) | 33.37 | (9.86) | 32.45 | (10.35) | **0.05** | **−2.01** | 0.59 | 0.54 |
| Age of diagnosis | 27.57 | (13.01) | 24.89 | (9.84) | 29.43 | (11.57) | 28.12 | (14.89) | 0.13 | −1.53 | 0.57 | 0.57 |
| Baseline ASRS | 49.97 | (8.53) | 48.72 | (8.86) | 49.26 | (9.16) | 50.98 | (7.98) | 0.09 | −1.71 | 0.22 | −1.24 |
| Baseline AAQoL | 41.87 | (13.19) | 42.58 | (13.65) | 41.63 | (12.80) | 41.61 | (13.23) | 0.65 | 0.46 | 0.99 | 0.01 |
| Treatment expectations | 6.26 | (1.67) | 5.84 | (1.58) | 6.20 | (1.79) | 6.51 | (1.61) | **0.01** | **−2.66** | 0.26 | −1.12 |

| Dichotomous variables | N | (%) | n | (%) | n | (%) | n | (%) | *p* | $Chi^2$ | *p* | $Chi^2$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Depression | 49 | (21.49%) | 13 | (21.31%) | 12 | (22.22%) | 24 | (21.24%) | 0.99 | 0.00 | 0.89 | 0.02 |
| Anxiety | 138 | (60.53%) | 33 | (54.10%) | 40 | (74.07%) | 65 | (57.52%) | 0.66 | 0.19 | **0.04** | **4.29** |
| Personality pathology | 31 | (13.60%) | 8 | (13.11%) | 6 | (11.11%) | 17 | (15.04%) | 0.73 | 0.12 | 0.49 | 0.48 |
| Sex (female) | 166 | (73.13%) | 40 | (65.57%) | 41 | (75.93%) | 85 | (75.22%) | 0.18 | 1.82 | 0.76 | 0.09 |
| Attained higher education | 99 | (43.42%) | 17 | (27.87%) | 34 | (62.96%) | 48 | (42.48%) | 0.06 | 3.61 | **0.01** | **6.14** |
| Currently employed | 156 | (68.42%) | 44 | (72.13%) | 37 | (68.52%) | 75 | (66.37%) | 0.44 | 0.61 | 0.78 | 0.08 |
| Relationships | 154 | (67.54%) | 37 | (60.66%) | 39 | (72.22%) | 78 | (69.03%) | 0.27 | 1.24 | 0.67 | 0.18 |
| Currently medicated | 142 | (62.28%) | 35 | (57.38%) | 36 | (66.67%) | 71 | (62.83%) | 0.48 | 0.50 | 0.63 | 0.23 |
| Self-referred | 49 | (21.49%) | 11 | (18.03%) | 13 | (24.07%) | 25 | (22.12%) | 0.53 | 0.40 | 0.78 | 0.08 |

Note. Early dropout: participants who only completed the baseline assessment at T0; Late dropout: participants who completed assessment at baseline (T0) and mid-treatment (T1) but not post treatment (T2); Completers: participants who completed all assessments; Early/completers: comparison between early dropouts and completers; Late/completers: comparison between late completers and completers; M: mean; SD: standard deviation; *p*: *p*-value; *t*: independent samples *t*-test test statistic; $Chi^2$: Chi-square test statistic: text in bold indicate significance value below 0.05.

perspective. Discrepant data points were coded as missing to avoid undue influence on later analyses.

### 4.3. Adherence and dropout

#### 4.3.1. Treatment adherence

Out of the 228 patients included in the study, 121 (53.1%) completed 6 or more treatment modules, which, given that module 7 is a summary module, constitutes completing the full course of intervention. A minority of participants (13.6%) did not progress beyond the first treatment module. The mean number of modules completed across the sample was 4.8 out of a total of 7 modules ($SD = 2.4$).

#### 4.3.2. Study protocol adherence

Out of the 228 included participants, 113 completed all three assessments (study completers), while 54 completed assessments at T0 and T1 (late dropouts) and 61 completed only the baseline assessment at T0 (early dropouts).

As can be seen from Table 1, Late dropouts were found to have higher rates of comorbid anxiety than study completers. Furthermore, late dropouts were more likely than study completers to have attained a university education. Beyond this these participants did not differ significantly from study completers on any other baseline measure. Early dropouts, however, were found to be somewhat younger, and to have lower treatment expectations than study completers.

For the primary outcomes of ASRS and AAQoL total scores, two additional rounds of linear mixed models exploring change over time were fitted, representing: a) Study completers only; and b) All participants with missing values imputed using multiple imputation for monotone missing. Comparing findings across analyses revealed only minor differences in estimated pre-post effect sizes, with both completers only (ASRS d = −0.51; AAQoL d = 0.48) and imputation-based models (ASRS d = −0.47; AAQoL d = 0.49) estimating equal or greater effect sizes than the primary analysis (ASRS d = −0.47; AAQoL d = 0.46). These results were taken to indicate limited impact of missing data and attrition on change estimates. To prevent overestimating treatment effects, the primary analysis was carried forward and subsequent sub-group and predictor analyses were estimated using random intercept, fixed slopes linear mixed models using all available data.

### 4.4. Primary and secondary outcomes

Table 2 outlines results from the linear mixed models estimated to examine change over time across primary (total score) and secondary (subscale scores) measures of ADHD symptoms (ASRS) and quality of life (AAQoL). These analyses indicate moderate pre-post change in both ADHD symptom severity (d = − 0.47) and quality of life (d = 0.45). Subscale analyses for ASRS indicate greater pre-post improvement of inattention (d = − 0.46) than hyperactivity (d = −0.35), although both are significant. Similarly, for quality of life, subscale analyses indicate the greatest pre-post improvement to the sub-scale productivity (d = 0.46) and relationships (d = 0.43) and the least improvement to life outlook (d = 0.19). Interestingly, improvements to ASRS total score and subscale scores were significant both from pre-treatment to mid-treatment and from mid-treatment to post-treatment. For quality of life on the other hand, the improvements were primarily significant from pre-to-mid treatment with only the productivity sub-scale showing continued improvement beyond mid-treatment.

### 4.5. Absolute change, reliable change and clinical significance

Study completers ($N = 113$) exhibited an average *absolute change* of −4.46 on the ASRS (SD = 6.51), and + 6.83 on the AAQoL (SD = 12.99). Calculating individual percentage change from pre to post treatment-completers exhibited, on average an 8.75% reduction (SD = 13.03) in ADHD symptoms (ASRS) and a 22.20% increase (SD = 42.36) in quality of life (AAQoL).

Following the formula by Jacobson and Truax (1991), reliable change was calculated for both ASRS and AAQoL. For ASRS, change in excess of ±8.85 indicated reliable change. Following this threshold, 23.89% of study completers exhibited a reliable decrease in ADHD symptoms from pre- to post-treatment. Two participants (1.77%) exhibited a reliable increase in ADHD symptoms from pre- to post-treatment. For AAQoL, change in excess of ±13.68 indicated reliable change. Following this threshold, 30.97% of completers exhibited reliable improvements in quality of life. Five participants (4.42%) exhibited a reliable reduction in quality of life over the course of the treatment.

*Clinically meaningful change* was evaluated using established and proxy thresholds. Following minimal clinically important improvement (MCID) thresholds put forth by Tanaka et al. (2019), participants were classified as having achieved clinically meaningful improvement in quality of life if their AAQoL scores increased by more than 8 points. In the absence of an established raw-score MCID threshold for the ASRS, a reduction of 30% or more in symptom severity was used as a default threshold for clinically meaningful change in ADHD symptoms. Based on these criteria, 45.13% of study completers were classified as responders with respect to quality-of-life. In contrast, only 4.42% of participants met criteria for clinically meaningful reduction in ADHD symptoms.

### 4.6. Predictor analysis

Table 3 summarizes results of predictor analyses for the two primary outcome measures of ASRS and AAQoL total scores. As can be seen from

**Table 2**
Overview of results from the linear mixed model analyses of change over time across primary and secondary outcomes.

| | Pretreatment (T0) | | Week 3 of treatment (T1) | | Post treatment (T2) | | Pre to mid | | | | Mid to post | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | M (SD) | | M (SD) | | M (SD) | | b | (95% CI) | *p* | d | b | (95% CI) | *p* | d |
| *ASRS total score* | 49.97 | (9.01) | 48.10 | (6.92) | 45.77 | (8.08) | −1.87 | (±0.90) | 0.00* | −0.21 | −2.33 | (±1.05) | 0.00* | −0.26 |
| ASRS inattention | 26.56 | (4.66) | 25.79 | (4.25) | 24.40 | (4.99) | −0.77 | (±0.55) | 0.01* | −0.16 | −1.39 | (±0.65) | 0.00* | −0.30 |
| ASRS hyperactivity/ impulsivity | 23.41 | (6.04) | 22.26 | (3.91) | 21.29 | (4.56) | −1.15 | (±0.51) | 0.00* | −0.19 | −0.97 | (±0.59) | 0.00* | −0.16 |
| *AAQoL total score* | 41.87 | (13.84) | 46.65 | (12.58) | 48.13 | (14.86) | 4.78 | (±1.63) | 0.00* | 0.35 | 1.48 | (±1.93) | 0.13 | 0.11 |
| AAQoL productivity | 38.49 | (16.25) | 42.80 | (15.12) | 46.01 | (17.86) | 4.31 | (±1.96) | 0.00* | 0.27 | 3.21 | (±2.32) | 0.01* | 0.20 |
| AAQoL psychological health | 37.48 | (19.05) | 44.49 | (17.82) | 43.52 | (21.07) | 7.01 | (±2.31) | 0.00* | 0.37 | −0.97 | (±2.73) | 0.49 | −0.05 |
| AAQoL relationships | 46.89 | (19.47) | 53.39 | (20.44) | 55.25 | (24.24) | 6.51 | (±2.65) | 0.00* | 0.33 | 1.86 | (±3.15) | 0.25 | 0.10 |
| AAQoL life outlook | 47.38 | (15.15) | 49.65 | (14.84) | 50.27 | (17.56) | 2.27 | (±1.93) | 0.02 | 0.15 | 0.62 | (±2.28) | 0.60 | 0.04 |

Note. N with no missing values: (T0 = 228, T1 = 167, T2 = 113); ASRS: Adult ADHD Self Report Scale; AAQoL: Adult ADHD Quality of Life Questionnaire; b: regression coefficient (beta); 95% CI: 1.96 * SE to indicate distance to upper and lower bound of 95% confidence interval; d: Cohen's d calculated from differences between time points; M: mean; SD: standard deviation.

* Significance value ≤ 0.01 marked for reader-convenience.

**Table 3**
Predictor analyses for pre-post change in ASRS and AAQoL total scores.

| | | *ASRS total score* | | | | | | *AAQoL total score* | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Baseline (95% CI) | | *p*-Diff | Pre-post change (95% CI) | | *p*-Diff | Baseline (95% CI) | | *p*-Diff | Pre-post change (95% CI) | | *p*-Diff |
| Age | Mean | 49.97 | (±1.17) | 0.08 | −4.22 | (±1.04) | 0.76 | 41.87 | (±1.80) | 0.10 | 6.21 | (±1.90) | 0.20 |
| | Per unit | 0.11 | (±0.12) | | 0.02 | (±0.10) | | −0.16 | (±0.19) | | 0.12 | (±0.19) | |
| Sex | Female | 51.04 | (±1.35) | 0.00* | −3.93 | (±1.20) | 0.33 | 40.52 | (±2.09) | 0.01* | 5.85 | (±2.20) | 0.44 |
| | Male | −3.90 | (±2.60) | | −1.20 | (±2.40) | | 5.14 | (±4.02) | | 1.75 | (±4.40) | |
| Employment status | Not employed | 50.47 | (±2.09) | 0.57 | −3.94 | (±1.79) | 0.72 | 38.78 | (±3.16) | 0.02 | 4.83 | (±3.29) | 0.29 |
| | Employed/student | −0.74 | (±2.52) | | −0.40 | (±2.20) | | 4.52 | (±3.82) | | 2.19 | (±4.03) | |
| Education level | Not university educated | 49.60 | (±1.56) | 0.48 | −4.07 | (±1.38) | 0.75 | 42.73 | (±2.39) | 0.29 | 5.35 | (±2.53) | 0.27 |
| | University educated | 0.86 | (±2.36) | | −0.35 | (±2.10) | | −1.97 | (±3.63) | | 2.16 | (±3.84) | |
| Expectation | Mean | 49.97 | (±1.17) | 0.48 | −4.23 | (±1.05) | 0.82 | 41.87 | (±1.80) | 0.81 | 6.21 | (±1.92) | 0.75 |
| | Per unit | 0.26 | (±0.70) | | 0.07 | (±0.64) | | 0.13 | (±1.08) | | 0.20 | (±1.18) | |
| Referral pathway | Externally referred | 49.67 | (±1.32) | 0.34 | −4.03 | (±1.18) | 0.54 | 42.36 | (±2.03) | 0.31 | 5.45 | (±2.15) | 0.12 |
| | Self-referred | 1.39 | (±2.85) | | −0.79 | (±2.50) | | −2.27 | (±4.38) | | 3.65 | (±4.56) | |
| Medication status | Not medicated | 49.51 | (±1.91) | 0.55 | −4.80 | (±1.71) | 0.39 | 40.75 | (±2.93) | 0.34 | 7.52 | (±3.12) | 0.32 |
| | Medicated | 0.74 | (±2.42) | | 0.94 | (±2.15) | | 1.81 | (±3.71) | | −2.02 | (±3.93) | |
| Baseline depression | Not depressed | 49.49 | (±1.31) | 0.12 | −4.73 | (±1.16) | 0.06 | 44.96 | (±1.84) | 0.00* | 6.55 | (±2.12) | 0.57 |
| | Depressed | 2.25 | (±2.82) | | 2.47 | (±2.51) | | −14.37 | (±3.97) | | −1.35 | (±4.60) | |
| Baseline anxiety | Not anxious | 45.97 | (±1.73) | 0.00* | −4.85 | (±1.61) | 0.28 | 49.74 | (±2.58) | 0.00* | 5.88 | (±2.92) | 0.79 |
| | Anxious | 6.61 | (±2.22) | | 1.17 | (±2.10) | | −13.00 | (±3.32) | | 0.53 | (±3.83) | |
| Baseline PD | Not PD | 49.55 | (±1.25) | 0.08 | −4.30 | (±1.12) | 0.65 | 43.85 | (±1.79) | 0.00* | 6.74 | (±2.05) | 0.31 |
| | PD | 3.06 | (±3.38) | | 0.69 | (±2.92) | | −14.54 | (±4.86) | | −2.78 | (±5.35) | |

Note. ASRS: Adult ADHD Self Report Scale; AAQoL: Adult ADHD Quality of Life Questionnaire; Baseline: intercept and predictor main effect; Pre–post chance: main effect of time and interaction effect of time * predictor; 95% CI: 1.96 * SE to indicate distance to upper and lower bound of 95% confidence interval; *p*-Diff: *p*-value associated with predictor main effect and time * predictor interaction effect.
* Significance value ≤ 0.01 marked for reader-convenience.

this table, none of the included variables were found to be significant predictors of treatment change. The closest estimate being comorbid depression at baseline which was found to have a near-significant association with ASRS change, such that depressed participants showed less improvement in self-reported ADHD symptoms over the course of treatment.

At baseline however, there were substantial sub-group differences across both outcomes, with women and participants with comorbid anxiety having greater baseline ASRS scores than men and non-anxious participants. Similarly, women; participants with comorbid anxiety, depression or personality pathology; and participants who were unemployed had lower baseline quality of life. In particular, comorbid mental disorders were associated with substantially lower quality of life at baseline. Interestingly however, pre-post change in quality of life was unaffected by these baseline differences.

To account for the potential inter-relatedness of comorbidity, an additional set of post-hoc analysis were carried out in which random intercept, fixed slope linear mixed models including all mean-centered continuous measures of depression (PHQ-9), anxiety (GAD-7) and personality pathology (PDS-ICD-11) as separate fixed effects and interaction terms with time. These analyses returned similar results as that seen in the separate analyses described in Table 3; where greater scores on GAD-7 were found to be significantly associated with baseline ASRS and greater scores on GAD-7, PHQ-9 and PDS-ICD-11 were all found to be comparably related to decreased baseline AAQoL scores.

To account for the potential additive impact of multimorbidity, separate models were estimated including a continuous variable representing multimorbidity (0 = no comorbid disorders, 1 = one comorbid disorder, 2 = two comorbid disorders, 3 = three comorbid disorders). This multimorbidity variable was entered in as a separate fixed effect and as an interaction term with time. These models indicated that multimorbidity was associated with greater ASRS scores and lower AAQoL scores at baseline but not associated with differences in pre-post change.

## 5. Discussion

The current study explored pre-post change and potential predictors of change in adults with ADHD enrolled in a therapist-guided internet-delivered intervention provided in routine care. Overall, the study found small to moderate improvements in ADHD symptoms (d = 0.47) and quality of life (d = 0.45). This is in line with previous studies examining the effect of non-pharmacological interventions on ADHD (Liu et al., 2023; Ostinelli et al., 2025) and previous studies exploring internet-delivered interventions for adult ADHD (Kenter et al., 2023; Liu et al., 2024; Nasri et al., 2023). Given the side effects and adherence challenges associated with stimulant medications, the ready availability of an effective psychological treatment represents an important complement or alternative within routine care (Graham et al., 2011; Graham and Coghill, 2008; Silczuk et al., 2025).

In line with previous non-pharmacological intervention studies (Liu et al., 2023), our study found the greatest improvement to the domains of inattention and productivity, suggesting that the current intervention may be particularly suited to teaching compensatory strategies for attentional difficulties.

### 5.1. Overall response to treatment

Compared to the previous study by Kenter et al. (2023), the current study saw more modest effect sizes and a lower proportion of participants achieving reliable change. This discrepancy likely reflects differences in target population and sample characteristics. Kenter et al. (2023) recruited a self-selected, community-based sample that enrolled rapidly: within 48 hours of study announcement, suggesting high intrinsic motivation, digital literacy, and readiness for self-directed treatment. The current study on the other hand, sampled a naturalistic, clinical population, most of whom had been referred to treatment by healthcare professionals. While baseline ADHD symptom severity was comparable between the two samples, the current sample had lower baseline quality of life scores (baseline mean was 6.8 points lower in the current sample) and included fewer participants receiving pharmacological treatment for ADHD 62.3% in the current study versus 83.3% in Kenter et al. (2023)). While neither referral pathway, nor medication status were found to be significant predictors of treatment response in the current study, these differences between samples do underscore the differences in sample population between the two studies.

### 5.2. Clinically meaningful change and negative effects

While estimated effect sizes and reliable change estimates are comparable between measures of ADHD symptom severity and quality of life in the current study, estimates of meaningful clinical change differ substantially between outcomes. Whereas nearly half of study completers exhibited improvements in quality of life in excess of established MCID (Tanaka et al., 2019), very few participants showed meaningful improvements in ADHD symptom severity. Given the supposed stability and invariance of core ADHD symptoms however (Faraone et al., 2021), this is not necessarily a surprising finding in the context of non-pharmacological interventions.

Rates of deterioration, as indicated by the reliable change metric, was lower in the current sample than that of the intervention group in previous study by Kenter et al. (2023). Furthermore, it was substantially lower than what has previously been reported in internet-delivered interventions more broadly (Rozental et al., 2016). While this is encouraging, it is important to note that specific measures of patient reported negative effects (e.g., Rozental et al., 2019) were not actively assessed in the current study.

### 5.3. Predictor analysis

While the current study did not identify any significant predictors of pre-post change, we did identify multiple baseline differences in both severity of ADHD symptoms and quality of life across various subpopulations.

#### 5.3.1. Demographic factors

Female participants reported higher baseline ADHD symptom scores and lower quality of life. This is in line with previous studies suggesting that the threshold for being diagnosed as a woman is higher than that of males (Williamson and Johnston, 2015; Young et al., 2020). Interestingly, contrary to previous findings that women tend to be diagnosed later in life compared to men (Martin et al., 2024), there were no differences between the sexes in age of diagnosis in our sample.

#### 5.3.2. Contextual factors

None of the contextual factors explored were found to have any association with either baseline symptoms and quality of life, nor pre-post change. Of note here, is the observation that patients currently undergoing medical treatment for ADHD did not show lower baseline symptom scores or higher quality of life scores than unmedicated participants. This could reflect a selection effect, where more severe cases are more likely to be medicated, and medication reduces symptoms and improves quality of life to levels comparable with those of unmedicated individuals with less severe symptoms. Alternatively, while pharmacological treatment is generally considered the first line of recommended treatment, effect sizes do vary between studies, and combined treatment does not always outperform monotherapy (Corbisiero et al., 2018).

The observation that neither treatment expectation nor referral pathway predicted baseline scores, or pre–post change is encouraging. These findings suggest that, at least in the current sample, patients who may not initially consider digital intervention a first choice but who are willing to give it a try if suggested by their GP or therapist had similar effects of the current treatment as participants who actively sought out a digital intervention. These findings run contrary to that of previous studies that have found clear associations between treatment expectations and treatment response in both face-to-face and digital interventions (Constantino et al., 2018; Pontén et al., 2024). Furthermore, the finding of similar baseline characteristics between referred and self-referred patients in the current sample, is in line with previous studies suggesting that allowing for self-referral does not lead to a “treating the healthy” scenario where less severe patients who would not qualify for care elsewhere self-select to online interventions (Bjarke et al., 2025; Staples et al., 2022)

#### 5.3.3. Comorbidity

Comorbid psychopathology did not appear to negatively impact treatment response. This held true across both categorical and dimensional measures of depression, anxiety, and personality pathology. Together, these findings suggest that the presence of common comorbidities did not directly impair treatment effectiveness in the current sample. This is encouraging given that depression, anxiety and personality pathology are among the most common comorbid conditions to ADHD (Choi et al., 2022). It is important to note however, that the current study did not explore the impact of comorbidity on engagement, adherence and dropout. This should be further examined in future trials.

### 5.4. Limitations

Firstly, although the sample for the current study was large in the context of naturalistic non-pharmacological intervention studies, it is possible that the study was still underpowered to detect small differences in treatment response across predictor variables. This may be particularly the case for unevenly distributed dichotomous predictors such as comorbid depression and personality pathology. It is worth noting, however, that for the three comorbidities examined (anxiety, depression, personality pathology), analyses were repeated using continuous measures and null findings were replicated. Even so, minor differences in treatment response may have gone undetected, and the absence of evidence for interaction effects cannot necessarily be taken as evidence of absence.

Second, the current study relied exclusively on self-reported outcome measures, without the inclusion of clinician-rated, performance-based, or proxy-reported assessments. Incorporating additional measurement modalities would have strengthened the robustness of the findings. Relatedly, although we monitored deterioration based on changes in outcome measures, we did not administer specific questionnaires assessing negative effects or adverse events, which could have provided a more systematic understanding of potential harm.

Third, as a study conducted in a routine clinical setting at the eTreatment clinic at Haukeland University Hospital, Bergen, Norway, the findings may have limited generalizability to other service contexts. The clinic is a specialized outpatient service with extensive experience in delivering therapist-guided internet interventions. Settings without a similarly established infrastructure may not achieve comparable outcomes. Relatedly, while participants were primarily referred by general practitioners or therapists, the sample may nonetheless be biased toward individuals who are more motivated, digitally literate, or otherwise predisposed to engage with internet-delivered care. Highly educated women also appear overrepresented relative to the broader adult ADHD population, which may further affect generalizability. It should be noted however, that this being a large naturalistic sample, local sample characteristics may indicate characteristics of the broader target population who will receive this intervention if and when the treatment becomes more widely implemented nationally. As such, this may be a limitation related to the public appeal of the intervention/intervention format, rather than a limitation of the sampling strategy of the current study.

Fourth, we did not track changes in baseline characteristics (e.g., as medication status) over the course of treatment. This limits the interpretability of symptom improvements, particularly in the absence of a control group, as unmeasured changes in clinical status could partially account for observed outcomes.

Fifth, since the intended sampling population for the current study was patients who would, in a naturalistic clinical setting, be eligible for the MyADHD intervention, inclusion was largely the discretion of the clinician conducting the intake evaluation for a given patient. While this is a clear strength in terms of the naturalistic design of the study, it is also a limitation in that diagnostic status was not directly verified, potentially introducing diagnostic variability into the sample. As with

demographics noted above however, this is perhaps best considered an artifact of the clinical context being studied, rather than a limitation of the current sample.

Sixth, the study did not include a control group. As a single-arm open trial, it cannot rule out the possibility that improvements in ADHD symptoms and quality of life were partly attributable to non-specific factors such as expectancy effects, regression to the mean, or the natural course of symptoms. As such, current findings warrant further investigation and replication in randomized controlled trials with suitable control conditions. This could involve comparing MyADHD to an alternative intervention.

Finally, the study did not examine long-term outcomes. It remains unclear whether treatment gains are maintained over time.

## 6. Conclusion

The current study adds to the growing evidence supporting therapist-guided internet-delivered interventions for adults with ADHD. On a group level, we found small-to-moderate improvements to both core ADHD symptoms and quality of life measures, particularly with regards to inattention and productivity. While modest, the effect sizes seen in the current study are comparable to that of other non-pharmacological interventions for ADHD (Liu et al., 2023; Ostinelli et al., 2025). On an individual level, we found that almost half of participants (45.13%) achieved clinically meaningful improvements to quality of life, while only a small minority (4.42%) achieved clinically meaningful reductions in ADHD symptoms. Predictor analyses indicated no significant impact of demographic factors, contextual factors or clinical comorbidities on the effect of treatment in the current sample. It should be noted, however, that subtle differences may have been indetectable despite the relatively large sample included in the current study. While replication in controlled trials is needed, these findings provide preliminary indications that MyADHD could potentially have utility for its target audience and that baseline comorbidity, referral pathway and treatment expectation may not present strong counter-indications for the intervention. Future research should include large-scale multi-cite randomized controlled trials and include long-term outcomes, as well as potential predictors of engagement, adherence and dropout from a "what works for whom"-perspective.

## Ethics declaration

The current study was approved by the Regional Committee for MedicalResearch Ethics of Western Norway (Reference number: 664966). Informed consent was provided by all participants. The study was conducted in accordance with the Declaration of Helsinki for Ethical Research (World Medical Association, 2013).

## Declaration of competing interest

The authors declare that there is no conflict of interest associated with the current publication. The author(s) disclosed receipt of the following financial support for the research, authorship and/or publication of this article: This research was supported by a Center for Research and Innovation grant, awarded to the last author (Grant no. NFR 309264 by the Research Council of Norway.

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